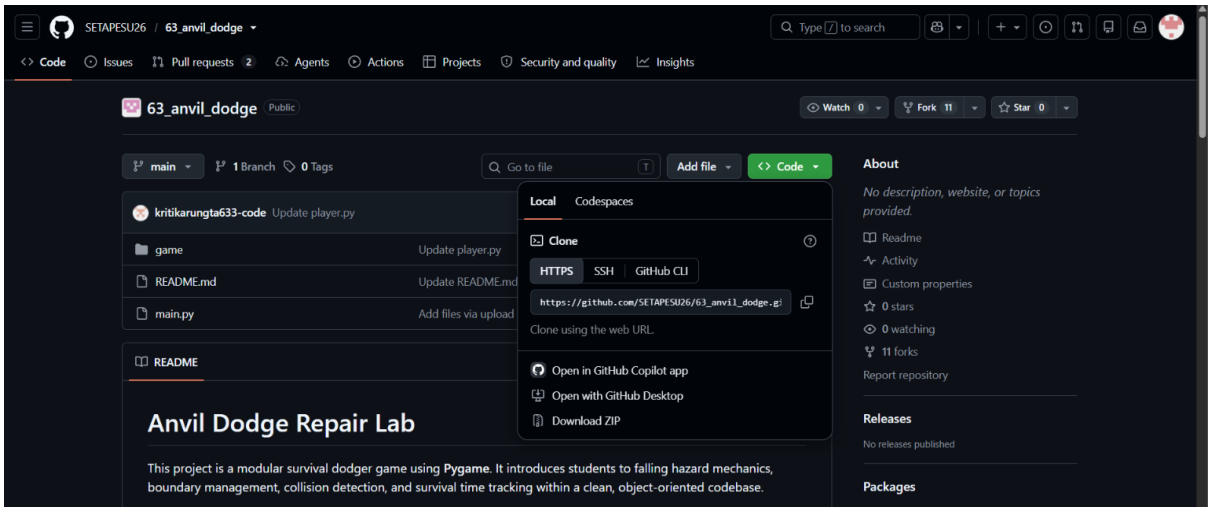
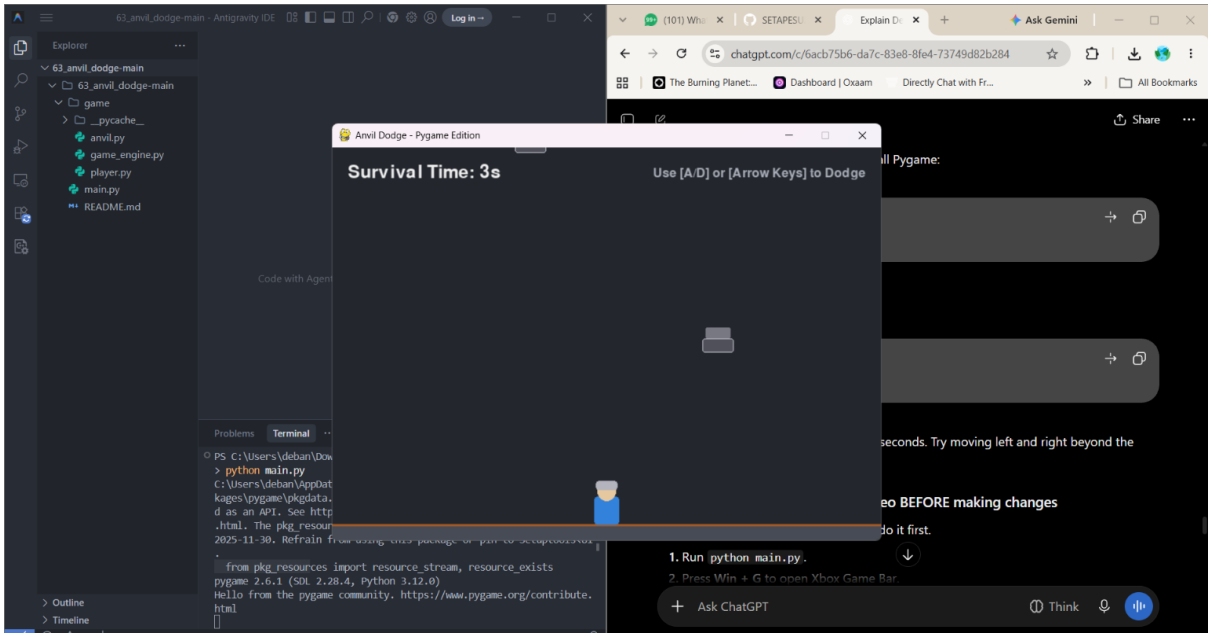
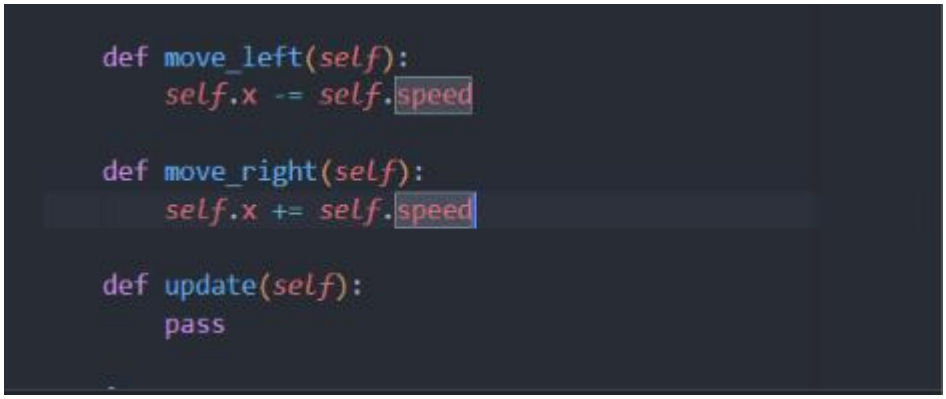
PES1UG25CS831	PRATISTHA MAJUMDER	SEC:H(5TH)





Task 1 — Player boundaries:

The bug in player.py



```
def move_left(self):
    self.x -= self.speed

def move_right(self):
    self.x += self.speed

def update(self):
    pass
```

The updating :

```
def move_left(self):
    self.x -= self.speed
    self.x = max(0, self.x)

def move_right(self):
    self.x += self.speed
    self.x = min(self.screen_width - self.width, s

    def update(self):
        pass
```

Task 2 — Dynamic difficulty:

Dynamic Difficulty Scaling in your Anvil Dodge game, step by step.

The goal is to make the game progressively harder as the player survives longer. We'll increase the anvil falling speed and gradually spawn anvils more frequently.

The fault in Engine.py:

```
        self.anvils = []
Ctrl+L for Agent
        self.spawn_delay = 700
        self.last_spawn_time = pygame.time.get_tic
```

Updated :

```
        # Dynamic difficulty settings
        self.base_spawn_delay = 700
        self.min_spawn_delay = 250
        self.spawn_delay = self.base_spawn_delay

        self.difficulty_interval = 5  # Increase di
        self.speed_increase = 0.5
        self.max_speed_bonus = 5

        self.last_spawn_time = pygame.time.get_tic
        self.start_ticks = pygame.time.get_ticks()
        self.survival_time = 0
        self.game_state = "PLAYING"
```

Task 3 — Speed-Based Anvil Warning Tints:

Before updating:

```
    def render(self, surface):
        top_rect = pygame.Rect(int(self.x) + 4, int(self.y), self.width - 8, 14)
        pygame.draw.rect(surface, (120, 120, 130), top_rect, border_radius=2)

        base_rect = pygame.Rect(int(self.x), int(self.y) + 14, self.width, 18)
        pygame.draw.rect(surface, (80, 80, 90), base_rect, border_radius=3)
        pygame.draw.rect(surface, (200, 200, 210), base_rect, width=1, border_radius=3)
```

After updating:

```
def render(self, surface):
    # Select warning colors based on anvil speed
    if self.speed >= 10:
        # Red: very fast anvil
        top_color = (255, 65, 65)
        base_color = (180, 25, 25)
        outline_color = (255, 150, 150)

    elif self.speed >= 8:
        # Orange: fast anvil
        top_color = (255, 165, 40)
        base_color = (190, 95, 20)
        outline_color = (255, 210, 130)

    else:
        # Grey: normal anvil
        top_color = (120, 120, 130)
        base_color = (80, 80, 90)
        outline_color = (200, 200, 210)
```

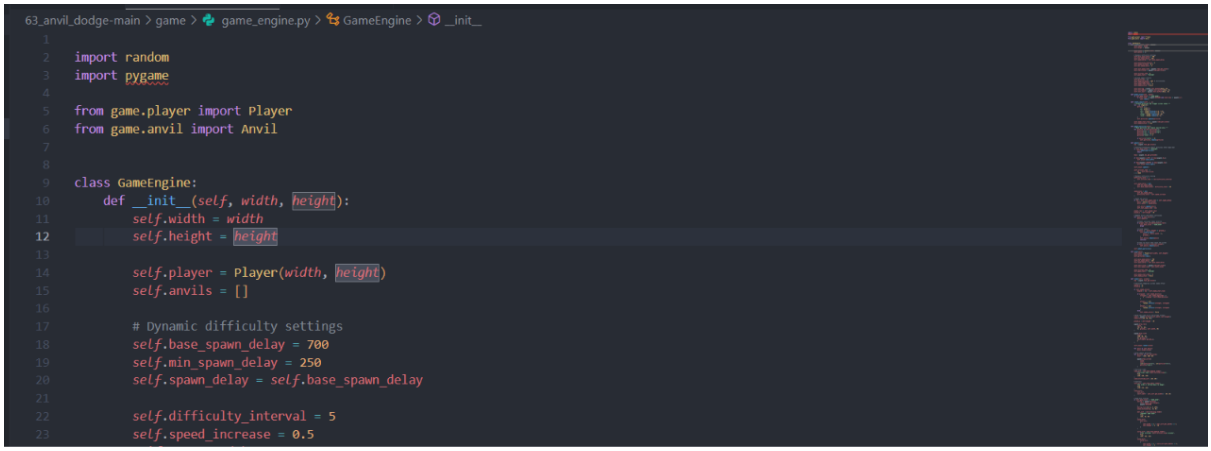
Task 4 — Ground Impact Effects and Screen Shake:

Before updating the engine.py

63_anvil_dodge-main > game > game_engine.py > GameEngine > __init__

```
1
2   import pygame
3   from game.player import Player
4   from game.anvil import Anvil
5
6
7   class GameEngine:
8       def __init__(self, width, height):
9           self.width = width
10          self.height = height
11          self.player = Player(width, height)
12          self.anvils = []
13
14          # Dynamic difficulty settings
15          self.base_spawn_delay = 700
16          self.min_spawn_delay = 250
17          self.spawn_delay = self.base_spawn_delay
18
19          self.difficulty_interval = 5  # Increase difficulty every 5 seconds
20          self.speed_increase = 0.5
21          self.max_speed_bonus = 5
22    Ctrl+L for Agent
23          self.last_spawn_time = pygame.time.get_ticks()
```

After updating:


63_anvil_dodge-main > game > game_engine.py > GameEngine > __init__
import random
import pygame
from game.player import Player
from game.anvil import Anvil
class GameEngine:
def __init__(self, width, height):
self.width = width
self.height = height
self.player = Player(width, height)
self.anvils = []
Dynamic difficulty settings
self.base_spawn_delay = 700
self.min_spawn_delay = 250
self.spawn_delay = self.base_spawn_delay
self.difficulty_interval = 5
self.speed_increase = 0.5